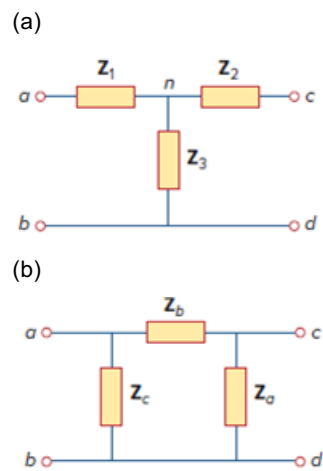


Problem

Assume that the two circuits in Fig. 19.135 are equivalent. The parameters of the two circuits must be equal. Using this factor and the z parameters, derive Eqs. (9.67) and (9.68).

Figure 19.135



Step-by-step solution

Step 1 of 8

Consider the following circuit diagrams:

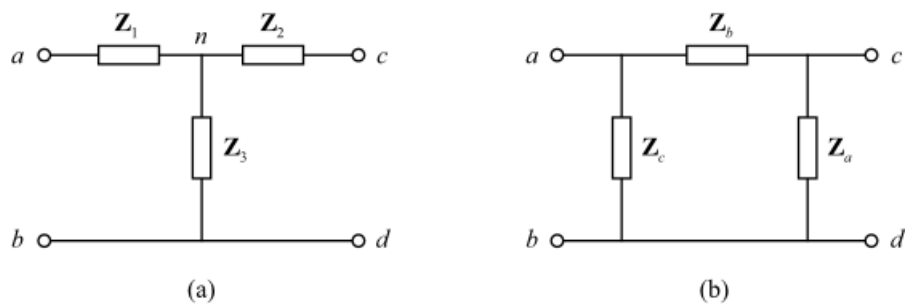


Figure 1

Step 2 of 8

Assume that the circuits (a) and (b) in Figure 1 are equivalent.

The parameters of the two circuits are equal.

The following is the input impedance of circuit (a).

$$Z_{ab} = Z_1 + Z_3$$

The following is the input impedance of circuit (b).

$$\begin{aligned} Z_{ab} &= Z_c \parallel (Z_b + Z_a) \\ &= \frac{Z_c (Z_b + Z_a)}{Z_a + Z_b + Z_c} \\ &= \frac{Z_b Z_c + Z_a Z_c}{Z_a + Z_b + Z_c} \end{aligned}$$

The input impedances of the circuits (a) and (b) in Figure 1 are equal.

$$Z_1 + Z_3 = \frac{Z_b Z_c + Z_a Z_c}{Z_a + Z_b + Z_c} \dots\dots (1)$$

Step 3 of 8

The following is the output impedance of circuit (a).

$$Z_{cd} = Z_2 + Z_3$$

The following is the output impedance of circuit (b).

$$\begin{aligned} Z_{cd} &= Z_a \parallel (Z_b + Z_c) \\ &= \frac{Z_a (Z_b + Z_c)}{Z_a + Z_b + Z_c} \\ &= \frac{Z_a Z_b + Z_a Z_c}{Z_a + Z_b + Z_c} \end{aligned}$$

The output impedances of the circuits (a) and (b) in Figure 1 are equal.

$$Z_2 + Z_3 = \frac{Z_a Z_b + Z_a Z_c}{Z_a + Z_b + Z_c} \dots\dots (2)$$

Step 4 of 8

The following is the impedance seen by terminals *a* and *c* of circuit (a).

$$Z_{ac} = Z_1 + Z_2$$

The following is the impedance seen by terminals *a* and *c* of circuit (b).

$$\begin{aligned} Z_{ac} &= Z_b \parallel (Z_c + Z_a) \\ &= \frac{Z_b (Z_c + Z_a)}{Z_a + Z_b + Z_c} \\ &= \frac{Z_b Z_c + Z_a Z_b}{Z_a + Z_b + Z_c} \end{aligned}$$

The impedances seen by terminals *a* and *c* of the circuits (a) and (b) in Figure 1 are equal.

$$Z_1 + Z_2 = \frac{Z_b Z_c + Z_a Z_b}{Z_a + Z_b + Z_c} \dots\dots (3)$$

Step 5 of 8

Subtract equation (2) from equation (1).

$$\begin{aligned} (Z_1 + Z_3) - (Z_2 + Z_3) &= \frac{Z_b Z_c + Z_a Z_c}{Z_a + Z_b + Z_c} - \frac{Z_a Z_b + Z_a Z_c}{Z_a + Z_b + Z_c} \\ Z_1 - Z_2 &= \frac{Z_b Z_c + Z_a Z_c - Z_a Z_b - Z_a Z_c}{Z_a + Z_b + Z_c} \\ Z_1 - Z_2 &= \frac{Z_b Z_c - Z_a Z_b}{Z_a + Z_b + Z_c} \dots\dots (4) \end{aligned}$$

Add equation (3) and (4) to get an expression for Z_1 .

$$\begin{aligned} Z_1 + Z_2 + Z_1 - Z_2 &= \frac{Z_b Z_c + Z_a Z_b}{Z_a + Z_b + Z_c} + \frac{Z_b Z_c - Z_a Z_b}{Z_a + Z_b + Z_c} \\ 2Z_1 &= \frac{Z_b Z_c + Z_a Z_b + Z_b Z_c - Z_a Z_b}{Z_a + Z_b + Z_c} \\ &= \frac{2Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_1 &= \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \dots\dots (5) \end{aligned}$$

Subtract equation (5) from equation (3) to get an expression for Z_2 .

$$\begin{aligned} Z_1 + Z_2 - Z_1 &= \frac{Z_b Z_c + Z_a Z_b}{Z_a + Z_b + Z_c} - \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_2 &= \frac{Z_b Z_c + Z_a Z_b - Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_2 &= \frac{Z_a Z_b}{Z_a + Z_b + Z_c} \dots\dots (6) \end{aligned}$$

Subtract equation (6) from equation (2) to get an expression for Z_3 .

$$\begin{aligned} Z_2 + Z_3 - Z_2 &= \frac{Z_a Z_b + Z_a Z_c}{Z_a + Z_b + Z_c} - \frac{Z_a Z_b}{Z_a + Z_b + Z_c} \\ Z_3 &= \frac{Z_a Z_b + Z_a Z_c - Z_a Z_b}{Z_a + Z_b + Z_c} \\ Z_3 &= \frac{Z_a Z_c}{Z_a + Z_b + Z_c} \dots\dots (7) \end{aligned}$$

Step 6 of 8

Use equations (5), (6), and (7) to calculate $Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1$.

$$\begin{aligned} Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1 &= \left[\left(\frac{Z_b Z_c}{Z_a + Z_b + Z_c} \right) \left(\frac{Z_a Z_b}{Z_a + Z_b + Z_c} \right) \right] \\ &\quad + \left[\left(\frac{Z_a Z_b}{Z_a + Z_b + Z_c} \right) \left(\frac{Z_a Z_c}{Z_a + Z_b + Z_c} \right) \right] \\ &\quad + \left[\left(\frac{Z_a Z_c}{Z_a + Z_b + Z_c} \right) \left(\frac{Z_b Z_c}{Z_a + Z_b + Z_c} \right) \right] \\ &= \frac{Z_a Z_b^2 Z_c}{(Z_a + Z_b + Z_c)^2} + \frac{Z_a^2 Z_b Z_c}{(Z_a + Z_b + Z_c)^2} + \frac{Z_a Z_b Z_c^2}{(Z_a + Z_b + Z_c)^2} \\ &= \frac{Z_a Z_b^2 Z_c + Z_a^2 Z_b Z_c + Z_a Z_b Z_c^2}{(Z_a + Z_b + Z_c)^2} \\ &= \frac{(Z_a Z_b Z_c)(Z_a + Z_b + Z_c)}{(Z_a + Z_b + Z_c)^2} \\ Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1 &= \frac{Z_a Z_b Z_c}{Z_a + Z_b + Z_c} \dots\dots (8) \end{aligned}$$

Divide equation (8) by equation (5) to get an expression for Z_a .

$$\begin{aligned}\frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1} &= \left(\frac{Z_a Z_b Z_c}{Z_a + Z_b + Z_c} \right) \\ &= \left(\frac{Z_b Z_c}{Z_a + Z_b + Z_c} \right) \\ &= \frac{Z_a Z_b Z_c}{Z_b Z_c} \\ &= Z_a\end{aligned}$$

Divide equation (8) by equation (6) to get an expression for Z_b .

$$\begin{aligned}\frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2} &= \left(\frac{Z_a Z_b Z_c}{Z_a + Z_b + Z_c} \right) \\ &= \left(\frac{Z_a Z_b}{Z_a + Z_b + Z_c} \right) \\ &= \frac{Z_a Z_b Z_c}{Z_a Z_b} \\ &= Z_c\end{aligned}$$

Divide equation (8) by equation (7) to get an expression for Z_c .

$$\begin{aligned}\frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3} &= \left(\frac{Z_a Z_b Z_c}{Z_a + Z_b + Z_c} \right) \\ &= \left(\frac{Z_a Z_c}{Z_a + Z_b + Z_c} \right) \\ &= \frac{Z_a Z_b Z_c}{Z_a Z_c} \\ &= Z_b\end{aligned}$$

Step 8 of 8

Thus, the following are the expressions for the z parameters of circuit (a).

$$\begin{aligned}Z_1 &= \frac{Z_b Z_c}{Z_a + Z_b + Z_c} \\ Z_2 &= \frac{Z_a Z_b}{Z_a + Z_b + Z_c} \\ Z_3 &= \frac{Z_a Z_c}{Z_a + Z_b + Z_c}\end{aligned}$$

The following are the expressions for the z parameters of circuit (b).

$$\begin{aligned}Z_a &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1} \\ Z_b &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3} \\ Z_c &= \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}\end{aligned}$$

Note: The expressions for the circuit parameters are not exactly the same as those in Chapter 9(Equations 9.67 and 9.68).

The location of Z_b and Z_c are different for the two situations.